

What are typical instruction cycle steps for the von-neuman architecture? describe.

- -fetch: get the next instruction from memory using the program counter.
- -Decode: figure out what the instructions mean and what needs to be done.
- -Execute: carry out the operation specified by the instructions using ALU.
- -Memory access: if needed interact with memory (read or write data).
- -Write back: save the result of the operation.
- -Update program counter: move to the next instruction in memory.

breaking logical memory into blocks of the same size called? paging

What is Round-Robin scheduling?

- Each process is assigned a fixed time slot or quantum. If a process does not complete within its time quantum, it is moved to the back of the queue, and the next process in the queue is given the CPU.

What is the shortest remaining time, SRT scheduling?

- is a preemptive scheduling, in SRT, the process with smallest runtime to complete is scheduled to run next, including new arrivals?

What is a CPU-I/O Burst Cycle?

- pattern where the CPU performs computations and then waits for input/output operations to finish. This cycle repeats as the program runs and interacts with external devices.

what are the 4 component of process?

- -Stack:
- -Heap: data structure that used to store a collection of elements. It's typically implemented as binary tree, where the parent's nodes are either smaller or bigger than chilled nodes depending on wither it is main heap or max heap.
- -Data section: region of memory in a program executable code where data values are stored, this data can include variables and other information that it be used by the program.

- -Text section: also known as code section, region of memory in a program executable where the instruction of the program are stored, it can only read one contains the machine code that is executed by the process.

What is swapping? moving processes temporarily out of memory to a backing store.

What is external fragmentation?

- When the available memory or disk space become scattered or broken into small pieces, making it difficult to find enough space for the new data or program, it happens in the stored system or OS.

What are the 2 fundamental models of inter process communication?

- shared memory:
 - is when process is a common memory space to share and exchange data directly which is quick and efficient, However, it needs careful coordination using tools like semaphores or locks to prevent issues.
- Message Passing:
 - Mechanism for processes to communicate and to synchronize their actions Message system - processes communicate with each other without resorting to shared variables IPC facility provides two operations: send(message) receive(message) The message size is either fixed or variable.

What is segmentation?

- It is a method in computer system for dividing a process memory into a smaller part called segment, this helps with better memory utilization, improved protection and clearer structure for the programmers.

What is the fundamental component of the process control Block?

- process state: the state may be new, ready, and running etc.
- program counter: the counter indicates the address of the next instruction to be executed for this process.

- CPU register: small, fast storage unit in the CPU used to hold data and instruction being processed and provide quick access to frequently used information, improving the efficiency of CPU operations.
- memory management information: memory allocated to the process.

What is the absolute and relative path?

- Absolute path starts from root directory whereas relative path starts from current directory.

What is the term used to describe the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order access? race condition.

What are the attributes of a file or directory?

- Name: only information kept in human-readable form.
- Identifier: unique tag (number) identifies file within file system.
- Type: needed for systems that support different types.
- Location: The location or path where the directory is stored in the file system.
- Size - current file size

What is internal fragmentation?

- condition in computer memory management, happened when there is a waste of memory due to the allocator returning a larger block than the memory requested.

What are the preemptive and non-preemptive strategies for process scheduling?

- Preemptive: The OS can interrupt a running process and replace it with another process, this helps to balance the processing load, such: round robin.
- non-preemptive: Once a process starts, it cannot be interrupted until it's done or waiting. This is simpler but can lead to starvation, such FCFS.

What is the term for describing the situation where awaiting process is never again able to change state, because the resources it has requested are held by other waiting process?

Deadlock

What is context switch?

- the state of one process, and restore the state of another

What kernel data structure can be used for one technique of passing parameters to system call?

- stack: liner data structures that follow the LIFO principal and it contain a temporary data such (local variable and return addresses).

just as a file must be opened before it is used, a file system must be **--mounted--** before it can be available to process on system.

what are the 3 requirement solution to the critical section problem , must satisfy?

- -Mutual Exclusion:
 - Ensures that only one process at a time can execute in its critical section to prevent concurrent access and potential data corruption.
- -Progress:
 - Guarantees that if no process is currently in its critical section and some processes are waiting, at least one of them will be granted access, promoting continuous execution.
- -Bounded Waiting:
 - Limits the number of times other processes can enter their critical sections before a waiting process is granted access, preventing indefinite postponement of a process.

What is the advantage of multi programming?

- increase CPU utilization: allow CPU to switch between programs.
- increase responsiveness: reduce response time for each process.
- increase flexibility: allow different types to run.

- better using experience: allow user to run multiple programs at the same time.

What is the general disk scheduling algorithms?

- FCFS: first come first serve.
- SSTF: shortest seek time first.
- scan: start at one end and move to another, reversing back when it gets to each end.
- C-scan (Circular SCAN): instead of reversing it moves from one end to the other, then immediately proceeds back to the end it started scanning from
- Look

What is paging?

- A memory management technique used by OS to efficiently manage the available physical memory by dividing it into small chunks called pages, Only the necessary are loaded into memory when a process needs them , allowing for efficient use of memory.

what is the term used to describe the segment of code where shared data is accessed and possibly manipulated? **critical section.**

Describe the critical section problem with your own example.

- several people want to fish in the same spot, However if they all start fishing in the same time the fishing line might get tangled and cause a problem. To avoid that, everyone should wait till the first fisherman finish and leave , then the second man take his place and so on .This way each person will wait his turn , preventing interference and ensuring a smooth and enjoyable fishing experience for everyone.

What is the dual mode operation of the OS?

- -User mode:
 - where regular applications run.
 - limited access to avoid messing with critical system stuff.

- -Kernel mode (super visor mode):
 - reserved for the OS.
 - has a higher privilege and can access everything.
- This dual mode setup protects the system. User applications cannot easily mess with critical parts. When an app needs special access, it asks the OS (using the system call) to do it in kernel mode, This way, the system stays secure and stable.

What is a system call?

- An interface to the services made available by an operating system.

Describe the term "LINK" in the file system contest.: **(A pointer to a existing file There are two types of link Hard link and soft link).**

what is reader-writer problem?

- a synchronization problem that occurs when multiple processes want to access a shared resource simultaneously. The goal is to allow multiple readers to access the resource at the same time but with only one writer, to avoid conflicting access.

a program that has been loaded and executed is called? **Process**

what is the problem if all philosophers simultaneously pick up their left (fork / chopsticks) ?

- it will leave no right (fork / chopsticks) for them to pick up, creating a deadlock while they all indefinitely wait for a right (fork / chopsticks) to be available .

what are the 3 strategies for selecting a free hole from the set of available holes ?

- -First fit : allocate the first hole that is big enough.
- Best fit : allocate the smallest hole that is big enough.
- Worst fit : allocate the largest hole that is big enough.

explain semaphores with details.

- semaphores are a synchronization tool with two types –
 - Binary semaphore
 - Counting semaphore

Binary semaphore can take the value 0 & 1 only. Counting semaphore can take nonnegative integer values. Two standard operations, wait and signal are defined on the semaphore. Entry to the critical section is controlled by the wait operation and exit from a critical region is taken care of by signal operation.

```
typedef struct {
    int value;
    struct process *list;
} semaphore;

wait (semaphore *S) {
    S->value--;
    if (S->value < 0) {
        add this process to S->list;
        block();
    }
}

signal(semaphore *S) {
    S->value++;
    if (S->value <= 0) {
        remove a process P from S->list;
        wakeup(P);
    }
}
```

Explain microkernel architecture with details.

- Microkernel architecture is a design approach for operating systems where the core kernel is kept minimal, handling essential tasks such as inter-process communication and basic hardware abstraction. Non-essential functions like file systems and device drivers are implemented as separate user-space processes. This modular design enhances flexibility, reliability, and security by isolating components and allowing for easy updates.

What is signal handling?

- a way for programs to respond to events like errors or interruptions. When a signal occurs, processes can be programmed to take specific actions, such as stopping gracefully or ignoring the signal. It helps manage events during program execution.

What is turnaround time refers to?

- the sum of the periods spent waiting to get into memory, waiting in the ready queue, executing on the CPU, and doing I/O.

How to make a Round Robin scheduling algorithm behave like (FCFS)?

- Remove Time Limits:
 - In Round Robin, each task has a time limit. Make this time limit very large or infinite so that tasks run until they finish.
- Keep Order:
 - Unlike Round Robin, don't reshuffle tasks based on time limits. Let them stay in the order they came in.
- No Interruptions:
 - In Round Robin, tasks can be interrupted after a certain time. In the modified version, don't interrupt tasks; let each one finish before the next starts.

What is Peterson's?

- Peterson's is a method in computer programming to make sure that only one process can use a shared resource at a time. It's like a rulebook for two processes, helping them take turns accessing a critical section of code. This avoids conflicts and ensures that shared resources are used properly.

do {

```
flag[i] = true;
```

```
turn = j;
```

```
while (flag[j] && turn == j);
```

critical section

```
flag[i] = false;
```

remainder section

```
} while (true);
```
